

API Testing

An API (Application Programming Interface) is a set of rules and protocols that allows different software applications to communicate, exchange data with each other. It acts as a third person to send and receive data. This provides seamless integration between software applications that furthermore improves the experience of the user. It is important because:

- Connects different software systems
- Enable app to fetch data from servers
- Used in web apps, IoT devices, cloud services

API testing is a software testing type that validates Application Programming Interface (API). Its purpose is to check functionality, reliability, performance and security of the programming interface.

REST vs SOAP APIs

REST	SOAP
Stands for Representational State Transfer	Stands for Simple Object Access Protocol
Uses file format as JSON, XML, HTML	Uses only XML
Uses HTTP/HTTPS	Uses HTTP, SMTP, TCP
Easier and lightweight	More complex and heavyweight
Flexible and scalable	More rigid than REST
Especially used in modern web & mobile apps	Used in enterprise-grade apps

JSON

JavaScript Object Notation (JSON) is a lightweight data format used to represent structured data. It is a simple text format and is easy to transmit. It is used in APIs, configuration files and data exchange between servers and clients. JSON used key-value pairs which makes it easier to understand.

JSON	XML
Human-readable	Human-readable but more wordy
Smaller and more compact	Larger due to extra markup
Easier to parse in most languages	More complex parsing
Broad support across languages	Initially JavaScript, but now widely supported
Used in Web APIs, configuration files, data transfer	Used in Data storage, document formatting